



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.RP.3A

Graph Ratio Tables

Lesson 3-3 • Teacher Copy

Name: _____

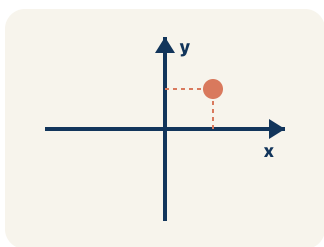
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

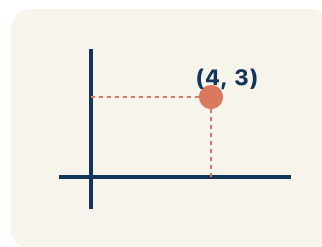
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A + shape made by two number lines crossing at the center point $(0, 0)$

Coordinate plane

Write the definition:



$(3, 6)$ means go right 3, up 6

Ordered pair

Write the definition:



$(1, 2)$, $(2, 4)$, $(3, 6)$ all line up

Linear pattern

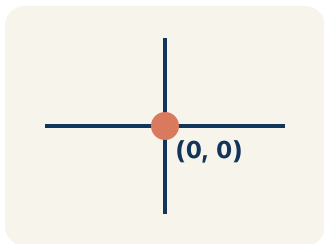
Write the definition:

$$\frac{1}{2} = \frac{2}{4}$$

A straight line from $(0, 0)$ through $(2, 6)$ and $(4, 12)$

Proportional

Write the definition:



The starting corner of the grid at $(0, 0)$

Origin

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can graph the values from a ratio table as points on the coordinate plane.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Coordinate plane

Ordered pair

Linear pattern

Proportional

Origin



Tap any word to see what it means and a picture.

1 The flat grid formed by a horizontal and a vertical number line is the

2 A set of two numbers (x, y) that names a point is an .

3 When graphed points form a straight line, they show a .

4 Two quantities that always keep the same ratio are .

5 The point (0, 0) where the two axes meet is the .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

- A. (8, 24)
- B. (7, 20)
- C. (8, 22)
- D. (10, 24)

Step 1 The pattern adds 2 to x and 6 to y each time.

Step 2 After (6, 18): $x = 6 + 2 = 8$, $y = 18 + 6 = 24$.

Step 3 The next point is (8, 24).

 **Answer:** A. (8, 24)

 Try — put the steps in order

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

The pattern adds 2 to x and 6 to y each time.

The next point is (8, 24).

After (6, 18): $x = 6 + 2 = 8$, $y = 18 + 6 = 24$.

 We do — together

Solve this one with your class using the same steps.

Problem: When you graph equivalent ratios, the points will always form what shape?

- A. A straight line through the origin
- B. A curved line
- C. A zigzag
- D. A circle

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when $x = 7$?

- A. 28
- B. 24
- C. 21
- D. 35

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The smoothie recipe is 1 cup of blueberries to 2 scoops of yogurt. If cups go on the x-axis and scoops on the y-axis, which ordered pair shows 3 cups of blueberries?

- A. (3, 6)
- B. (6, 3)
- C. (3, 5)
- D. (3, 3)

Show your work:

2. On the smoothie graph, every point follows the rule scoops = $2 \times$ cups. Which equation describes the pattern between x (cups) and y (scoops)?

- A. $y = 2x$
- B. $y = x + 2$
- C. $y = x \div 2$
- D. $x = 2y$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Show your work:

Reflect — Exit Ticket

A ratio table shows cups of rice to cups of water: (1, 2), (2, 4), (3, 6). If you plot these points, the line passes through which point?

- A. (5, 10)
- B. (4, 6)
- C. (5, 8)
- D. (6, 10)

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Linear pattern
4. **Notes 4:** Proportional
5. **Notes 5:** Origin
6. **We Try (notes):** A. A straight line through the origin — *Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).*
7. **You Try (notes):** A. 28 — *The ratio is $y/x = 8/2 = 4$. For every x , $y = 4x$. When $x = 7$: $y = 4 \times 7 = 28$.*
8. **Try It 1:** A. (3, 6) — *Cups (x) come first, scoops (y) second. 3 cups need $3 \times 2 = 6$ scoops, so the point is (3, 6).*
9. **Try It 2:** A. $y = 2x$ — *Each cup needs 2 scoops, so y (scoops) is always 2 times x (cups): $y = 2x$. For 4 cups, $y = 2 \times 4 = 8$.*
10. **Exit Ticket:** A. (5, 10) — *The ratio is 1:2, so for 5 cups of rice you need $5 \times 2 = 10$ cups of water. The point (5, 10) continues the pattern.*